



The Missing Block

Quick facilitation notes & answer key

Goal

Students fix a bug where a program falls short of the target because a block is missing; they add the missing step and run again to confirm. Altering existing code by adding a step (and writing that step) covers C3.2 and a focused piece of C3.1.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-5 so students see Bit stop short and count the steps left to the target.

Run it (about 20 min)

1. I DO: run Program 1 — 'start 1, forward 2 to 3. The target is 4 — Bit is one short. We need one more forward.'
2. WE DO: Exercise 1 — Bit lands on 3, one short. Exercise 2 — add 'forward 1'; the program reaches 4.
3. YOU DO: Exercise 3 — Program 2 lands on 3, two too high for target 1; add 'back 2' so $3 \rightarrow 1$.
4. Connect: 'Falling short can mean a missing block — add the step that finishes the job.'

Answer key (modelled by the run-gate)

Ex 1 — Program 1: $1 \rightarrow \text{forward } 2 \rightarrow 3$. Bit is on 3, one short of 4.

Ex 2 — add 'forward 1': $1 \rightarrow 3 \rightarrow 4$. Bit reaches the target, 4.

Ex 3 — Program 2: $4 \rightarrow \text{back } 1 \rightarrow 3$. Add 'back 2': $3 \rightarrow 1$. Bit reaches the target, 1. (Two 'back 1' blocks also work.)

Watch for & differentiate

Common slip: adding a block that overshoots — have students count the gap from where Bit stopped to the target, then add a step of that size.

Simplify: do Program 1 only (one forward short).

Extend: ask for a different way to fix Program 2 (e.g. two 'back 1' blocks) and test it.

Success indicator: the child adds 'forward 1' to reach 4 and a back block (back 2, or two back 1s) to reach 1.